

SECTION 16269
VARIABLE FREQUENCY DRIVES

PART 1 GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section includes: Solid-state, PWM, VFD's for speed control of squirrel-cage induction motors and all related accessories.

1.3 DEFINITIONS

- A. BMS: Building management system.
- B. IGBT: Insulated gate bipolar transistor.
- C. LAN: Local area network.
- D. PID: Control action, proportional plus integral plus derivative.
- E. PWM: Pulse-width modulated.
- F. ASD: Adjustable Speed Drive.
- G. VFD: Variable frequency drive. An ASD that uses frequency variation as the method of speed control.

1.4 SUBMITTALS

- A. Product Data: For each type of VFD. Include dimensions, mounting arrangements, location for conduit entries, shipping and operating weights, and manufacturer's technical data on features, performance, electrical ratings, characteristics, and finishes.
- B. Harmonic Analysis: Show compliance with voltage and current harmonic limits described in the Input Line Conditioning section below.
- C. Shop Drawings: For each VFD. See Division 16 "Electrical General Provisions".

1). Include dimensioned plans, elevations, sections, and details, including required clearances and service space around equipment. Show tabulations of installed devices, equipment features, and ratings. Include the following:

- a). Each installed unit's type and details.
- b). Nameplate legends.
- c). Short-circuit current rating of the VFD and size and type of any external "pre-fuses" that may be required to meet this short-circuit current rating.
- d). Space requirements for all enclosures required for housing the components specified, including but not limited to enclosures for the VFD, "pre-fuses", and harmonic filters.

2). Wiring Diagrams: Power, signal, and control wiring for VFD's. Provide schematic wiring diagram for each type of VFD.

D. Manufacturer's factory test reports.

E. Field quality-control test reports.

F. Operation and Maintenance Data: For VFD's, all installed devices, and components to include in emergency, operation, and maintenance manuals. In addition to items specified in Division 1, include the following:

1). Routine maintenance requirements for VFD's and all installed components.

2). See Division 16 "Electrical General Provisions".

1.5 QUALITY ASSURANCE

~~Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, Article 100, by a testing agency acceptable to authorities having jurisdiction, and marked for intended use.~~

~~Comply with NFPA 70.~~

~~Comply with UL 508C.~~

~~Comply with NEMA ICS 61800-2.~~

1.141.6 DELIVERY, STORAGE, AND HANDLING

- A. Store VFD's indoors in clean, dry space with uniform temperature to prevent condensation. Protect VFD's from exposure to dirt, fumes, water, corrosive substances, and physical damage.

~~1.151.7~~ PROJECT CONDITIONS

- A. Environmental Limitations: Rate equipment for continuous operation, capable of driving full load in the environment that it is installed. See Division 16 Section "General Electrical Provisions."

~~1.161.8~~ COORDINATION

- A. Coordinate layout and installation of VFD's with other construction including conduit, piping, equipment, and adjacent surfaces. Maintain required workspace clearances and required clearances for equipment access doors and panels.
- B. Where concrete bases are shown on drawings, coordinate size and location with other trades. See Division 16 "Basic Electrical Material and Methods".
- C. Where VFD's are installed on roofs, coordinate installation of roof curbs, equipment supports, and roof penetrations. These items are specified in Division 7 Section "Roof Accessories."
- D. Coordinate features, accessories, and functions of each VFD and each installed unit with ratings and characteristics of supply circuit, motor, required control sequence, and duty cycle of motor and load.

PART 2 PRODUCTS

2.1 MANUFACTURERS

- ~~A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:~~

- ~~1). Danfoss Inc.; Danfoss Electronic Drives Div.~~
- ~~— Eaton Corporation~~
- ~~2). General Electric~~
- ~~2). Toshiba International Corporation.~~
- ~~2). Yaskawa~~

2.2 VARIABLE FREQUENCY DRIVES

~~A. Description: A solid state controller configured to provide variable speed control of an NEMA MG 1, Design B, and induction motor by adjusting output voltage and frequency.~~

~~0). Input rectifier: Full wave diode bridge.~~

~~0). Output Inverter: IGBT, PWM type - 6, 12, or 18 pulse configuration as necessary to meet harmonic requirements in the Input Line Conditioning section below.~~

G.A. Design and Rating: Match motor and load combinations intended to be controlled by a VFD, including but not limited to; full load amperage, horsepower, or kW of motor, duty cycle, and variable or constant torque load.

H.B. Output Rating:

- 1). Variable torque applications: 6 to 60 Hz, with voltage proportional to frequency throughout voltage range.
- 2). Constant torque applications: 6 to 60 Hz, with torque constant as speed changes.

I.C. Unit Operating Requirements:

- 1). Input ac voltage tolerance of 460V plus or minus 10 percent.
- 2). Input frequency tolerance of 60 Hz, plus or minus 2 percent.
- 3). Minimum Efficiency: 95 percent at full load and full speed (60Hz).
- 4). Minimum Displacement Primary-Side Power Factor: 95 percent lagging over the entire speed range.
- 5). Overload Capability:
 - a). Normal overload (variable torque) applications: 110% for 1 minute.
 - b). High overload (constant torque) applications: 150% for 1 minute.
- 6). Starting Torque: 100 percent of rated torque or as indicated.
- 7). Speed Regulation: Plus or minus 1 percent. Motor slip dependent speed regulation.
- 8). The drive shall permit switching between local and remote operation as well as between manual and auto "on the fly".

- 9). Automatic Restart: Attempts a minimum of three restarts on return of power after an interruption. Number of restart attempts shall be selectable.
- 10). Catch "on the fly": Bidirectional auto-speed search shall be capable of starting into rotating loads spinning in either direction and returning motor to set speed in proper direction, without damage to controller, motor, or load.
- 11). Torque Boost: Automatically varies starting and continuous torque to at least 1.5 times the minimum torque to ensure high-starting torque and increased torque at slow speeds.
 - a). Dwell time at start adjustable from 0.5 to 10 seconds to maximize starting torque.
 - b). Low frequency boost up to 60 volts at 2-15 hertz.
- 12). Motor Temperature Compensation at Slow Speeds: Adjustable current fall-back based on output frequency for temperature protection of self-cooled, fan-ventilated motors at slow speeds.
- 13). Emergency shutdown of the controller which immediately disconnects the motor.

J.D. Internal Adjustability Capabilities:

- 1). Minimum and Maximum Speeds: Independently adjustable through the entire frequency range specified in the Output Rating section above (10 to 100 percent of maximum rpm).
- 2). Acceleration and Deceleration: Independently adjustable, in a linear or S-curve pattern, from 2 to 60 seconds for the frequency range specified in the Output Rating section above.
- 3). Current Limit:
 - a). Variable torque applications: Adjustable from 50 to 110 percent of drive's maximum output current rating.
 - b). Constant torque applications: Adjustable from 50 to 150 percent of drive's maximum output current rating.
- 4). Volts per hertz: Configurable to a Variable or Constant Torque pattern.
- 5). Carrier frequency: Pulse width adjustable from 1 to 5 kHz, minimum, in 1kHz maximum increments.

K.E. Self-Protection and Reliability Features:

- 1). Input transient protection by means of surge suppressors. Adequate to protect the drive over the entire range of the drive operation range.
- 2). Under and over-voltage trips; inverter over-temperature, overload, and over-current trips.
- 3). Upon failure of a cooling fan, the VFD shall initiate a remote warning indication at an adjustable temperature setting below the inverter over-temperature fault level.
- 4). Motor thermal overload protection: Solid state, speed dependant, adjustable and capable of NEMA 250, Class 10, 20, and 30 performance.
- 5). Critical (skip) frequencies: Notch filter to prevent operation of the controller-motor-load combination at a natural frequency of the combination. Provide a minimum of three programmable settings with adjustable bandwidths.
- 6). Instantaneous line-to-line and line-to-ground over-current trips.
- 7). Loss-of-phase protection.
- 8). Unaffected by input line phase rotation/phase sequence.
- 9). Short-circuit protection: Ability to withstand available input and output terminal line-to-line and line-to-ground short circuits without component failure (UL 508 Type 2).
 - a). 65kAIC minimum unless indicated otherwise on the drawings.
 - b). Provide fast acting semiconductor type fuses on the incoming AC line, or if external pre-fuses are required to meet short circuit rating, provide them as well as any cabinet required to house them as part of the VFD package.
- 10). Ride-through: Control logic shall have 2 second minimum ride through.

L.F. Harmonic Mitigation:

- 1). DC bus (link) reactor (choke): Series connected inductors in the positive and negative DC buses.
 - a). Provide 3% DC bus reactors for all drives.

- 2). AC line reactor: Series connected inductors in the incoming AC line.

~~— Provide AC line reactors sized at 3% of the line impedance for all 50HP and 75HP larger motors.~~

~~a). TCI Sine Guard KLR Series or approved equal.~~

- ~~2). Harmonics filter: Series connected inductors in the incoming AC line and shunt connected inductors and capacitors, tuned below the lowest harmonic generated by the drive.~~

~~a). Provide tuned LC filter for all 100HP and larger motors.~~

~~a). Include manual and automatic control feature to disconnect filter from system while motor is not running.~~

~~a). TCI Harmonic Guard Trap Filters or approved equal.~~

M.G. VFD Output Filtering:

- 1). dV/dT Filter: A low-pass, RLC filter, combination of series connected inductors with shunt connected resistors and capacitors connected at the output of the VFD.
 - a). Provide dV/dT filter for multi-motor applications.
 - b). Provide dV/dT filter for applications with motor lead lengths over 350ft unless other lengths are documented as acceptable by the VFD manufacturer. Provide documentation of acceptable motor lead lengths, based on a 5kHz carrier frequency, for owner approval prior to ordering equipment.

N.H. Key Pad: Backlit LCD multifunction keypad, flush mounted in VFD door.

- 1). Full operating, monitoring, and programming functionality.
- 2). Replaceable while VFD is under power without affecting motor.
- 3). Local/Remote and Manual/Auto selection keys.
- 4). Start key.
- 5). Stop/Reset key.
- 6). Parameter value/menu scroll keys.
- 7). Historical Logging Information and Displays:
 - a). Real-time clock with current time and date.

- b). Running log of total power versus time.
- c). Total run time.
- d). Fault log, maintaining last four faults with time and date stamp for each.
- e). Power savings.

O.I. Control Signal Interface:

~~1).~~ ~~Electric Input Signal Interface: A minimum of 2 analog inputs (0 to 10 V or 0/4-20 mA) and 6 programmable digital inputs.~~

~~3).~~ 1). Remote Signal Inputs: Capability to accept any of the following speed-setting input signals:

- a). 0 to 10-V dc.
- b). 0-20 or 4-20 mA.
- c). Potentiometer using up/down digital inputs.
- d). Fixed frequencies using digital inputs.
- e). RS485.
- f). Keypad display for local hand operation.

~~4).~~ 2). Output Signal Interface:

a). A minimum of 1 analog output signal (0/4-20 mA), which can be programmed to any of the following:

- 1. Output frequency (Hz).
- 2. Output current (load).
- 3. DC-link voltage (VDC).
- 4. Motor torque (percent).
- 5. Motor speed (rpm).
- 6. Set-point frequency (Hz).

~~5).~~ ~~Remote Indication Interface: A minimum of 2 dry circuit relay outputs (120-V ac, 1 A) for remote indication of the following:~~

- ~~).~~ ~~Motor running.~~
- ~~).~~ ~~Set point speed reached.~~
- ~~).~~ ~~Fault and warning indication (over temperature or over current).~~
- ~~).~~ ~~PID high or low speed limits reached.~~

~~12).~~ 3). Controller status relay with four contacts, each rated 3 amps resistive at 115 volts AC, for tripped, ready to run, up-to-speed, running, and forward/reverse indication. Each contact can be used as normally open or normally closed.

~~P-J.~~ Communications: Provide an RS485 interface allowing VFD to be remotely monitored and controlled.

~~Integral Disconnecting Means: NEMA KS 1, non fusible switch with lockable handle.~~

2.3 ENCLOSURES

~~A.~~ The VFD shall be NEMA 250, Type 1, unless otherwise required by the environment that is installed in, and shall be completely front accessible.

~~B-A.~~ Environment controls: refer to the Project Conditions section above.

~~1).~~ The VFD shall have cooling fans and air filters that are designed for easy replacement. The fans shall be designed for replacement without removing the VFD from the wall or removal of circuit boards. The cooling fans shall be controlled by the VFD to cycle on only as required to maintain a safe operating temperature.

~~0).~~

~~B.~~ Outdoor: Fans and/or heaters, as required to maintain the environment that the VFD is rated for without de-rating VFD.

~~C.~~ Integral 120 VAC power supply: The Drive shall require no external power source for controls, fans, heaters, or other accessories.

格式化: 大綱編號 + 階層: 3 + 編號樣式: 1, 2, 3, ... + 起始號碼: 1
+ 對齊方式: 左 + 對齊: 1.9 公分 + 定位點之後: 2.92 公分 + 縮排: 2.92 公分

格式化: 大綱編號 + 階層: 2 + 編號樣式: A, B, C, ... + 起始號碼: 1
+ 對齊方式: 左 + 對齊: 0.63 公分 + 定位點之後: 1.52 公分 + 縮排: 1.52 公分

格式化: 間距 套用前: 12 點

2.4 FACTORY FINISHES

A. Finish: Manufacturer's standard paint applied to factory-assembled and -tested VFD's before shipping.

2.5 FACTORY TESTS AND CHECKS

~~A.~~ All VFD's shall be tested by the manufacturer per NEMA ICS 61800-2.

~~G-A.~~ All features and accessories shall be functionally tested at the factory for proper operation.

~~D-B.~~ Report test results in writing.

PART 3 EXECUTION

3.1 INSTALLATION

A. See Division 16 Section "Basic Electrical Materials and Methods" for general installation requirements.

- B. Mount plumb and rigid. Provide all necessary blocking, channels and other hardware for securing VFD's to wall, column or other parts of building structure, according to manufacturer's written instructions.
- C. Provide equipment base for free standing VFD's. See Division 16 Section "Basic Materials and Methods."
- D. Minimum code required clearances must be maintained around VFD's. Clearance shall be adequate for doors to open a minimum of 90 degrees.

3.2 EQUIPMENT BASES

- A. Coordinate size and location of concrete bases. Verify structural requirements with structural engineer. See Division 16 Section "Basic Electrical Materials and Methods."

3.3 LABELING

- A. Label according to the connected load. Refer to drawings and Division 16 Section "Basic Electrical Materials and Methods."

3.4 CONTROL WIRING INSTALLATION

- A. Install wiring between VFD's and remote devices according to Division 16 Section "Conductors and Cables." Coordinate with Instrumentation and Controls.
- B. Bundle, train, and support wiring in enclosures.
- C. Connect hand-off-automatic switch and other automatic-control devices where applicable.

3.5 FIELD QUALITY CONTROL

- A. Prepare for acceptance tests as follows:
 - 1). Test insulation resistance for each enclosed controller element, bus, component, connecting supply, feeder, and control circuit.
 - 2). Test continuity of each circuit.
 - 3). Notify owner a minimum of 24 hours prior to energizing and/or testing VFD's.
- B. Manufacturer's Field Service: Engage a factory-authorized service representative to perform the following:

- 1). Inspect controllers, wiring, components, connections, and equipment installation.
 - 2). Assist in field testing of equipment including pre-testing and adjusting of solid-state controllers.
 - 3). Report final settings in writing.
- C. Measure harmonic content, voltage, and frequency at each drive's input and output terminals under full load conditions. Correct units that are out of spec and retest to demonstrate compliance. Report results in writing.

3.6 DEMONSTRATION

- A. Engage a factory-authorized service representative to train Owner's maintenance personnel to adjust, operate, and maintain variable frequency controllers. Training shall occur at owner's selected location.

END OF SECTION